Pose Detection

1. Experiment Objective

Use MediaPipe Pose to detect the 33 body pose keypoints in the camera image and visualize the human skeleton in real time.

2. Experiment Principle

1. Key Terms

Keyword	Description
MediaPipe Pose	Google's body pose keypoint detection model that predicts 33 3D body joint coordinates from a single frame
33 body keypoints	Covers the face (11 points), torso (4 points) and limbs (18 points); finer than OpenPose's 18 points
Keypoint smoothing	`smooth_landmarks=True` enables cross-frame smoothing to reduce keypoint jitter
Detection/tracking confidence	The minimum confidence thresholds for the initial detection stage and the ongoing tracking stage (default 0.5 each)

2. Technical Architecture

```
Terminal 1: start agent + camera + joint models
Terminal 2: start pose detection
```

3. Experiment Steps

1. Hardware Preparation

Hardware	Requirement
PTZ version	☑ Supported
No-PTZ version	☑ Supported
Required peripherals	ESP32 camera

2. Start the agent + camera + joint models (Terminal 1)

```
ros2 launch microros_v2_bringup car_base.launch.py
```

```
yahboom@yahboom:~$ ros2 launch microros_v2_bringup car_base.launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-07-28-10-47-55-372666-yahboom-149737
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [micro_ros_agent-1]: process started with pid [149778]
[INFO] [camera_driver-2]: process started with pid [149780]
[INFO] [robot_state_publisher-3]: process started with pid [149782]
[INFO] [joint_state_publisher-4]: process started with pid [149784]
[INFO] [ekf_node-5]: process started with pid [149803]
[camera_driver-2] [INFO] [1785206876.886150022] [esp32_ip_camera_publisher]: camera node started, camera_url: http://192.168.12.37:81/stream
[joint_state_publisher-4] [INFO] [1785206877.007751174] [joint_state_publisher]: Waiting for robot_description to be published on the robot_descriptio
[micro_ros_agent-1] [1785206877.173080] info     | UDPv4AgentLinux.cpp | init                     | running...             | port: 8090
[robot_state_publisher-3] [INFO] [1785206877.754391885] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-3] [INFO] [1785206877.754890084] [robot_state_publisher]: got segment base_link
[robot_state_publisher-3] [INFO] [1785206877.754912957] [robot_state_publisher]: got segment bwheel_Link
[robot_state_publisher-3] [INFO] [1785206877.754918130] [robot_state_publisher]: got segment cam1_Link
[robot_state_publisher-3] [INFO] [1785206877.754924428] [robot_state_publisher]: got segment cam2_Link
[robot_state_publisher-3] [INFO] [1785206877.754929199] [robot_state_publisher]: got segment cam3_Link
[robot_state_publisher-3] [INFO] [1785206877.754935071] [robot_state_publisher]: got segment imu_frame
[robot_state_publisher-3] [INFO] [1785206877.754939792] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-3] [INFO] [1785206877.754944035] [robot_state_publisher]: got segment lfwheel_Link
[robot_state_publisher-3] [INFO] [1785206877.754949951] [robot_state_publisher]: got segment rfwheel_Link
[camera_driver-2] [WARN] [1785206879.871708745] [esp32_ip_camera_publisher]: Camera not opened, trying reconnect... (next retry in 1.0s)
[camera_driver-2] [INFO] [1785206880.046238399] [esp32_ip_camera_publisher]: IP camera stream opened successfully
```

3. Start pose detection (Terminal 2)

```
ros2 launch microros_v2_mediapipe_pkg pose_detector.launch.py
```

4. Operating Instructions

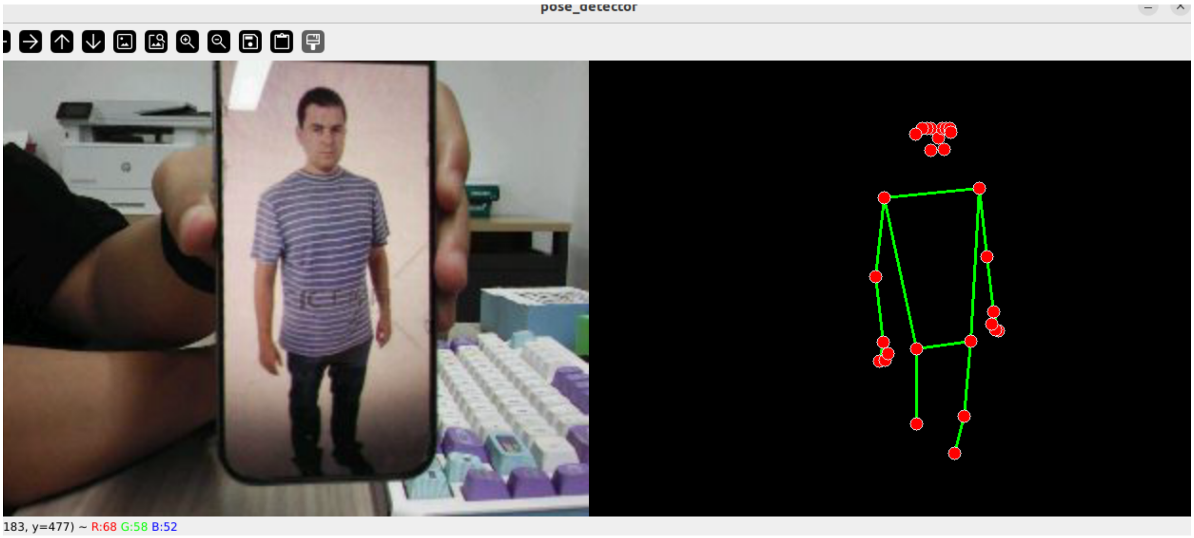
Press `q` to exit.

5. How to Stop

Press `Ctrl+C` in each terminal, one by one, to stop.

4. Observed Results

- 33 keypoints are marked at the corresponding joints (including 11 face points), with colored skeleton connections overlaid in real time
- With `smooth_landmarks=true`, keypoints show no obvious jitter between frames



5. Code Walkthrough

> Source code paths:
>
> - Launch: `~/microros_ws/src/microros_v2_mediapipe_pkg/launch/pose_detector.launch.py`

> - Node: `~/microros_ws/src/microros_v2_mediapipe_pkg/microros_v2_mediapipe_pkg/PoseDetector.py`

1. Core Node Analysis

```
# Source file:
~/microros_ws/src/microros_v2_mediapipe_pkg/microros_v2_mediapipe_pkg/PoseDetect
or.py
class PoseDetector(Node):
    def __init__(self):
        super().__init__("pose_detector_node")
        self.declare_parameter("rgb_topic", "/camera/image_raw")
        self.declare_parameter("static_image_mode", False)
        self.declare_parameter("smooth_landmarks", True)
        self.declare_parameter("detection_con", 0.5)
        self.declare_parameter("track_con", 0.5)

        self.mp_pose = mp.solutions.pose
        self.pose = self.mp_pose.Pose(
            static_image_mode=static_image_mode,
            smooth_landmarks=smooth_landmarks,
            min_detection_confidence=detection_con,
            min_tracking_confidence=track_con)

    def image_callback(self, msg):
        frame = self.bridge.imgmsg_to_cv2(msg, 'bgr8')
        frame_rgb = cv.cvtColor(frame, cv.COLOR_BGR2RGB)
        results = self.pose.process(frame_rgb)
        if results.pose_landmarks:
            self.mp_draw.draw_landmarks(
                frame, results.pose_landmarks, self.mp_pose.POSE_CONNECTIONS)
        cv.imshow(self.window_name, frame)
```

Key logic:

- `mp.solutions.pose.Pose()`: 33-keypoint detector
- `smooth_landmarks=True` (default): enables cross-frame smoothing to reduce keypoint jitter
- `static_image_mode=False`: enables cross-frame tracking, reducing computation in video stream mode
- `POSE_CONNECTIONS`: MediaPipe's predefined human skeleton connections, automatically drawing the skeleton lines

2. Key Parameters

> Parameter definition files:
>
> - Launch: `~/microros_ws/src/microros_v2_mediapipe_pkg/launch/pose_detector.launch.py`
> - Node: `~/microros_ws/src/microros_v2_mediapipe_pkg/microros_v2_mediapipe_pkg/PoseDetector.py`

Parameter	Type	Default	Description
`static_image_mode`	bool	false	Whether to detect each frame independently
`smooth_landmarks`	bool	true	Whether to enable cross-frame keypoint smoothing
`detection_con`	float	0.5	Detection confidence threshold
`track_con`	float	0.5	Tracking confidence threshold

3. Node Description

Node	Function
`pose_detector_node`	MediaPipe Pose 33-keypoint detection + skeleton visualization

6. Frequently Asked Questions

Issue	Cause	Solution
Keypoints jitter severely	`smooth_landmarks` is not enabled	Set `smooth_landmarks:=true`
Person not detected	Too far away or insufficient lighting	Bring the person closer to the camera